Comparative Metabolic Analysis of Fasting Effects on an EMT Model*

Exploring Metabolic Adaptations in EMT Models under Fasting Conditions.

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Abstract

This study employs metabolic modeling to investigate the impact of fasting on an Epithelial-Mesenchymal Transition model. Using constraint-based modeling techniques, we compare the metabolic flux distributions between an original model and a fasting integrated version for fasting conditions. Flux Variability Analysis is conducted to assess the range of feasible flux values for each reaction in both models, highlighting reactions significantly altered by fasting. This computational approach provides insights into metabolic adaptations during EMT under fasting conditions, potentially uncovering metabolic targets.

Load both the models and compare the growth rates

```
import cobra
from cobra.flux_analysis import flux_variability_analysis

# Load the original and modified models
original_model = cobra.io.read_sbml_model("""/Users/karthik/Desktop/
                                            PHCCO IISc Internship/EMT/
                                            MODEL1602080000_url.xml""")

integrated_model = cobra.io.read_sbml_model("""/Users/karthik/Desktop/
                                              PHCCO IISc Internship/EMT/
                                              integrated_MODEL1602080000.xml""")

# Compare growth rates
original_solution = original_model.optimize()
modified_solution = integrated_model.optimize()
```

*Thanks

```
print("Original model growth rate:", original_solution.objective_value)
print("Modified model growth rate:", modified_solution.objective_value)
```

```
Original model growth rate: 0.10099999998857717
Modified model growth rate: 0.10099999998857717
```

Perform FVA

- To understand the flexibility of each metabolic reaction by determining the maximum and minimum possible flux values (flux range) it can carry under steady-state conditions and given constraints.
- Before and after adjusting reaction bounds using functions like adjust_bounds, FVA can validate whether the adjusted constraints allow the reactions to operate within feasible physiological bounds.
- Identifies reactions that are critical (essential) or flexible (non-essential), which can further help in prioritising the reactions.

```
# Perform FVA on both models
original_fva = flux_variability_analysis(original_model)
modified_fva = flux_variability_analysis(integrated_model)

print("FVA results for original model:")
print(original_fva)
print("FVA results for modified model:")
print(modified_fva)
```

```
FVA results for original model:
                                 minimum       maximum
10FTHF7GLUtl                0.000000e+00   1666.666667
10FTHF7GLUtm                0.000000e+00   1666.666667
10FTHFtl                   -1.666667e+03      0.000000
10FTHFtm                   -1.000000e+04   3284.464975
13DAMPPOX                   6.568857e-14   1197.358346
...                                  ...           ...
NCAMUP                      0.000000e+00  10000.000000
NCAMDe                      0.000000e+00  10000.000000
sink_ncam_LSQBKT_c_RSQBKT_ -6.521726e-02      0.019581
EX_pnto_R_LPAREN_e_RPAREN_ -3.171588e-01     -0.049167
DM_pnto_R                   4.916697e-02      0.317159

[2853 rows x 2 columns]
FVA results for modified model:
                                 minimum       maximum
10FTHF7GLUtl                0.000000e+00   1666.666667
10FTHF7GLUtm                0.000000e+00   1666.666667
10FTHFtl                   -1.666667e+03      0.000000
10FTHFtm                   -1.000000e+04   3284.464975
```

```
13DAMPPOX                   6.568857e-14   1197.358346
...                                  ...           ...
NCAMUP                      0.000000e+00  10000.000000
NCAMDe                      0.000000e+00  10000.000000
sink_ncam_LSQBKT_c_RSQBKT_ -6.521726e-02      0.019581
EX_pnto_R_LPAREN_e_RPAREN_ -3.171588e-01     -0.049167
DM_pnto_R                   4.916697e-02      0.317159

[2853 rows x 2 columns]
```

Affected reactions based on change in Flux

```python
# Identify significantly affected reactions
affected_reactions = []

for rxn in integrated_model.reactions:
    if original_solution.fluxes[rxn.id] != modified_solution.fluxes[rxn.id]:
        affected_reactions.append(rxn.id)

print("Significantly affected reactions due to fasting:")
print(affected_reactions)
```

```
Significantly affected reactions due to fasting:
[]
```